

Rongfeng Cheng | Medical AI Research

Guangzhou, China

✉ eb1287945452@mail.scut.edu.cn • 🌐 crf2004.github.io/en • 📄 CRF2004
0009-0002-3021-9647

Research Focus

Research interests center on trustworthy and evidence-grounded medical AI: understanding whether models rely on clinically meaningful evidence, how evidence use changes across datasets, and how physiological signals can be aligned with language. Current work spans ECG model auditing, dataset-label evidence characterization, and ECG-text representation learning. Two first-author manuscripts are under review at IEEE JBHI and IEEE BIBM 2026.

Education

South China University of Technology (SCUT)

Guangzhou, China

B.S. in Big Data Management and Application

2023–2027

GPA: 3.46/4.0.

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Research Experience

First Author

Evidence-Use Auditing for Deep ECG Models

Jan 2026–Present

- Addressed whether high-performing ECG classifiers actually depend on clinically relevant signal evidence rather than task-correlated shortcuts.
- Developed a matched-control perturbation framework that compares clinically targeted interventions against matched control perturbations across evidence windows and perturbation doses.
- Evaluated the framework on PTB-XL and CODE-15% across model architectures and frozen-weight cross-dataset transfer; manuscript under review at *IEEE Journal of Biomedical and Health Informatics (JBHI)*.

First Author

DLEF: Dataset-Label Evidence Fingerprints

Jun 2026–Present

- Investigated whether dataset-specific evidence structure can explain why ECG classifiers lose performance when transferred across datasets.
- Proposed directional and boundary-aware evidence fingerprints and designed analyses relating these fingerprints to cross-dataset AUROC degradation.
- Evaluated CODE-15%, PTB-XL, and Chapman-Shaoxing; manuscript under review at IEEE BIBM 2026.

Researcher

PathCLIP: Evidence-Aware ECG-Text Representation Learning

Jul 2026–Present

- Investigating how diagnostic knowledge and ECG findings can provide clinically meaningful supervision for ECG-text alignment and zero-shot recognition.
- Designed matched-count, permuted-hierarchy, residual-only, and held-out-label controls to distinguish genuine knowledge gains from additional supervision or representation capacity.
- Current work shifts from ontology-only supervision toward finding-level evidence alignment.

Manuscripts & Publications

First author: R. Cheng et al. A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification. Under review at IEEE JBHI.

First author: R. Cheng et al. DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification. Under review at IEEE BIBM 2026.

Co-author: CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation. *iScience*, 2026. doi:10.1016/j.isci.2026.115073.

Co-author: Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques. HNNDL 2026. doi:10.1145/3795892.3795917.

Co-author: Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification. HNNDL 2026. doi:10.1145/3795892.3795918.

Professional Experience

AI Research Institute, Guangdong No. 2 Provincial People's Hospital

Guangzhou, China

Research Intern

Nov 2024–Present

- Work across three parallel areas in a hospital research environment: medical LLMs and clinical question answering, cardiovascular knowledge systems, and clinician-facing AI deployment.
- Built components of a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval; also contributed to a myocardial-ischemia knowledge graph.
- Developed clinical AI prototypes from workflow design through deployment, including the Dingbei Meimei acne and facial-health assessment system; built its full stack and iPad workflow and co-designed assessment scales with dermatologists for the July 2026 launch of the hospital's Refractory Acne Diagnosis and Treatment Center.
- Produced translational outputs across these workstreams, including a first-inventor patent application and software copyrights for medical Q&A assurance, knowledge-graph management, and LLM-based cardiovascular health management.

Selected Projects

Clinical NLP & Knowledge Systems

Medical LLM / Knowledge Representation

KG-Based Standardized Patient Q&A

Processed approximately 8,000 cardiovascular case PDFs and built an LLM-based extraction pipeline with named-entity recognition, relation extraction, entity disambiguation, Neo4j storage, and hybrid RAG.

Project Lead

Myocardial Ischemia Knowledge Graph

Built a domain knowledge graph with collaborative ontology management, XGBoost-based implicit reasoning, and graph-mapped interpretability analysis.

Clinical Data & Deployed Systems

Clinical AI / MIMIC-IV

IHD Trajectory Similarity Retrieval

Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays using time-windowed clinical variables and precomputed k-NN similarity graphs.

Full-Stack Developer; Assessment-Scale Co-Designer

Dingbei Meimei – Acne & Facial Health AI Assessment System

Built the complete front end and back end, including iPad adaptation and the clinical case workflow; co-designed the assessment scales with dermatologists at Guangdong No. 2 Provincial People's Hospital. Delivered the system for the July 2026 launch of the hospital's Refractory Acne Diagnosis and Treatment Center.

Patents & Intellectual Property

Patent: Method, System, and Electronic Device for Medical Question-Answering Output Assurance – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

Software IP: Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

Honors

- First Prize, 2025 Challenge Cup (Jinan University Campus).
- Second Prize, 2025 9th Guangzhou Mathematical Modeling University League.
- Third Prize, 2025 National Mathematical Modeling Contest – Guangdong Division.

Skills

Modeling: PyTorch, XGBoost, representation learning, multimodal learning

Evaluation: Perturbation analysis, matched controls, cross-dataset evaluation

Biomedical Data: 12-lead ECG, MIMIC-IV / EHR cohorts, clinical NLP, knowledge graphs

Engineering: Python, TypeScript, SQL, Next.js, Flask, Neo4j, Git, Docker

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528